

Dataanalyse

Anxiety levels



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1. Introduction

Anxiety is an ever increasing issue with more and more people, according to an article by psychiatry.org from may of 2024 (Connors, 2024). We wanted to look into this more and analyze how anxiety levels could be affected by various factors on an individual. Using a dataset with a rich amount of factors like demographic, lifestyle, treatments and more by using analysis tools learned during this course we can gain a better understanding on how these factors impact the anxiety levels.

Our goal for this project is to identify the most significant factors that contribute to increasing anxiety levels and judge which treatments have the most effect in reducing anxiety. We aim to answer the research questions through statistical and machine learning methods, with the end goal of being able to provide significant insights which can be taken into action in this area. We hope the insights gained from this project can be helpful for healthcare providers, companies and policy makers in investing and supporting mental health more than it currently is.

2. CRISP-DM

To achieve our goals we have decided to use the CRISP-DM method, this methodology is based on a six phase process, which is continually iterating. This is the most well known and most used model when it comes to data mining. This model is used as a standardized way to handle data mining in research projects and other data science projects. (Wirth & Hipp, 2000)

In the beginning of the process we need to gain a business understanding of our data, clarifying objectives and identifying how the data can be used to support these objectives. This leads directly into the data understanding phase where we collected data, explored it and assessed its usability. These two initial phases lay the foundations for the project.

Following this, we have to do some data preparation to ensure that we get usable results. This involves cleaning the data and structuring it for modeling. This is a phase where we often had to go back and revisit in the modeling phase as new issues with the data structure appeared. As mentioned, modeling is the next phase, which is where we build our predictive models and begin to analyze the data results through visualizations and statistics.

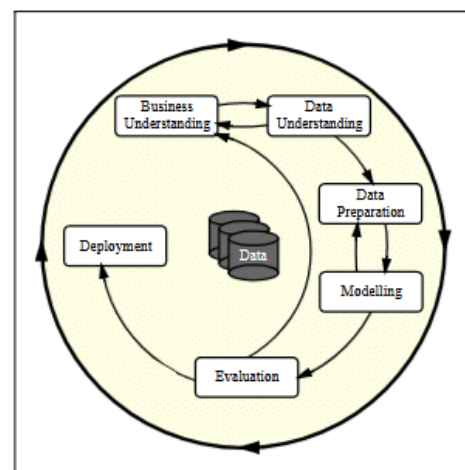


Figure 1: Phases of CRISP-DM (Wirth & Hipp, 2000)

When the models are done it is important to evaluate them to see if they meet the business objectives mentioned earlier and that they give relevant insights. Lastly, we have the deployment phase where we organize all of our findings and present them in a user friendly format. (Wirth & Hipp, 2000)

3. Business understanding

With the mentioned rise in anxiety we wanted to conduct this data-driven analysis of the different factors to help us in identifying individuals which are at a risk for high anxiety, which treatment types (therapy, medication etc.) are most effective in different types of patients and how external and internal factors can interact and influence the outcomes of mental health.

We hope the findings from this project can influence healthcare practices, influence corporate wellness programs and support more mental health research.

Problem Statement:

What are the key factors that influence anxiety levels, and how can they be used to predict or make treatment results better?

To get an answer for this we have these research questions:

Relationship between lifestyle factors and anxiety levels:

How do factors like caffeine intake, alcohol consumption, physical activity and sleep affect an individual's anxiety levels?

Familiar history and lifestyle effects on anxiety:

Does having a family history of anxiety have an increased chance of experiencing higher amounts of anxiety levels?

Major life events effect on anxiety levels:

How does major life events influence anxiety levels, and does this vary with different kinds of coping mechanisms, such as therapy or medication?

Physiological predictors for anxiety:

Can physical indicators (heart rate, breathing rate, dizziness) reliably predict anxiety level scores?

4. Data understanding and Data Preparation

In this phase, the goal was to gain a comprehensive overview of the dataset in order to identify relevant and irrelevant variables, as well as to uncover any need for data cleaning and further preparation. This is a crucial step before proceeding to data preprocessing and modeling.

We based our work on the following guiding questions:

1. Are the column names clear and meaningful, and what data types are present?
2. Are there any missing values, and in which columns?
3. Are there extreme or illogical values (outliers)?
4. Are there any duplicate entries or inconsistent/ambiguous values?
5. Which columns appear to be irrelevant for the analysis purpose?
6. Are any columns stored in the wrong format and need transformation before further analysis?

4.1. Data types and summary statistics

We examined data types and summary statistics using the “Statistics” view in RapidMiner to identify potential errors, inconsistencies, and the need for conversion. The column names were found to be understandable and meaningful. No missing values were detected, and no inconsistent or ambiguous entries were observed in categorical variables such as Gender, Smoking, or Family History of Anxiety.

To identify potential outliers, we focused on four numerical and continuous variables with a high potential for individual variation: Caffeine Intake, Alcohol Consumption, Heart Rate, and Breathing Rate. These variables are not limited to fixed scales, and may contain extreme values or recording errors.

We used the Normalize operator in RapidMiner with z-transformation to calculate the Z-score for each row in the four columns. We then used the Filter Examples operator to detect values where the Z-score exceeded ± 3 . No outliers were identified in the dataset.

To check for duplicate rows, we used the Aggregate operator with all columns selected under "Group by attributes" and counted occurrences using the count function. No rows were repeated, and we concluded that the dataset does not contain duplicates.

The variable Anxiety Level (1–10) was selected as the target variable in this project. Although it is stored as a numerical data type, it actually represents an **ordinal scale**—indicating levels rather than continuous quantities. RapidMiner interprets such values as numeric by default, which may lead to incorrect interpretation in models that expect categorical input.

Therefore, Anxiety Level **should be converted to a polynomial (categorical) datatype** before further modeling. This ensures that the values are treated as distinct classes rather than as numerical magnitudes, enabling more accurate interpretation in classification models.

Name	Type	Missing	Statistics			Filter (19 / 19 attributes): Search for Attributes
✓ 🚩 Age	Integer	0	Min 18	Max 64	Average 40.242	
✓ 🚩 Gender	Nominal	0	Least Other (3613)	Most Female (3730)	Values Female (3730), Male (3657), ...[1 more]	
✓ Occupation	Nominal	0	Least Teacher (807)	Most Musician (892)	Values Musician (892), Artist (888), ...[11 more]	
✓ Sleep Hours	Real	0	Min 2.300	Max 11.300	Average 6.651	
✓ Physical Activity (hrs/week)	Real	0	Min 0	Max 10.100	Average 2.942	
✓ Caffeine Intake (mg/day)	Integer	0	Min 0	Max 599	Average 286.090	
✓ Alcohol Consumption (drinks/w...)	Integer	0	Min 0	Max 19	Average 9.702	
✓ Smoking	Nominal	0	Least No (5221)	Most Yes (5779)	Values Yes (5779), No (5221)	
✓ Family History of Anxiety	Nominal	0	Least No (5153)	Most Yes (5847)	Values Yes (5847), No (5153)	
✓ Stress Level (1-10)	Integer	0	Min 1	Max 10	Average 5.856	
✓ Heart Rate (bpm)	Integer	0	Min 60	Max 119	Average 90.916	
✓ Breathing Rate (breaths/min)	Integer	0	Min 12	Max 29	Average 20.958	
✓ Sweating Level (1-5)	Integer	0	Min 1	Max 5	Average 3.081	
✓ Dizziness	Nominal	0	Least No (5328)	Most Yes (5672)	Values Yes (5672), No (5328)	
✓ Medication	Nominal	0	Least No (5334)	Most Yes (5666)	Values Yes (5666), No (5334)	
✓ Therapy Sessions (per month)	Integer	0	Min 0	Max 12	Average 2.428	
✓ Recent Major Life Event	Nominal	0	Least No (5377)	Most Yes (5623)	Values Yes (5623), No (5377)	
✓ Diet Quality (1-10)	Integer	0	Min 1	Max 10	Average 5.182	
✓ Anxiety Level (1-10)	Real	0	Min 1	Max 10	Average 3.929	

Figure 2: Data Types

4.2 Feature Engineering

The feature engineering process consisted of several steps to prepare the dataset for further analysis and modeling.

4.2.1 Feature Selection

To determine which variables were useful for predicting anxiety levels (which also addresses question 5), we applied several techniques. We used *Weight by Information Gain* to rank the importance of

predictors, a *correlation matrix* to explore linear relationships between numerical variables, and *box plots* to visualize the relationship between anxiety levels and categorical variables.

- **Weight by Information Gain:** We calculated information gain in RapidMiner relative to the target variable, *Anxiety Level*. The results showed that *Stress Level*, *Sleep Hours*, *Therapy Session*, *Caffeine Intake*, and *Diet Quality* were among the top five most important predictors. In contrast, *Gender*, *Medication*, and *Occupation* were found to be less relevant, and we decided to exclude them from further analysis and modeling.

attribute	weight
Gender	0.000
Medication	0.003
Occupation	0.007
Recent Major Life Event	0.008
Smoking	0.009
Dizziness	0.011
Family History of Anxiety	0.026
Alcohol Consumption (drinks/week)	0.035
Age	0.043
Sweating Level (1-5)	0.060
Breathing Rate (breaths/min)	0.066
Heart Rate (bpm)	0.080
Physical Activity (hrs/week)	0.081
Diet Quality (1-10)	0.103
Caffeine Intake (mg/day)	0.125
Therapy Sessions (per month)	0.193
Sleep Hours	0.208
Stress Level (1-10)	0.308

Figure 3: Weight by information gain

- **Correlation Matrix:** The correlation analysis confirmed that the variables with high information gain also showed stronger correlations -either positive or negative- with anxiety levels. Additionally, Physical Activity was identified as a relevant predictor.

Attribut...	Age	Sleep H...	Physica...	Caffein...	Alcohol...	Stress ...	Heart R...	Breathi...	Sweatin...	Therapy...	Diet Qu...	Anxiety ...
Age	1	-0.148	0.037	-0.038	-0.016	-0.045	-0.032	-0.014	-0.020	-0.087	0.049	-0.074
Sleep Ho...	-0.148	1	0.175	-0.209	-0.068	-0.177	-0.138	-0.120	-0.118	-0.309	0.154	-0.494
Physical ...	0.037	0.175	1	-0.115	-0.033	-0.103	-0.077	-0.071	-0.075	-0.188	0.086	-0.243
Caffeine ...	-0.038	-0.209	-0.115	1	0.036	0.121	0.077	0.080	0.076	0.217	-0.089	0.351
Alcohol ...	-0.016	-0.068	-0.033	0.036	1	0.052	0.044	0.024	0.024	0.062	-0.030	0.101
Stress L...	-0.045	-0.177	-0.103	0.121	0.052	1	0.088	0.063	0.085	0.209	-0.110	0.668
Heart Rat...	-0.032	-0.138	-0.077	0.077	0.044	0.088	1	0.053	0.062	0.155	-0.086	0.189
Breathing...	-0.014	-0.120	-0.071	0.080	0.024	0.063	0.053	1	0.048	0.139	-0.054	0.157
Sweating...	-0.020	-0.118	-0.075	0.076	0.024	0.085	0.062	0.048	1	0.118	-0.077	0.160
Therapy ...	-0.087	-0.309	-0.188	0.217	0.062	0.209	0.155	0.139	0.118	1	-0.173	0.518
Diet Qual...	0.049	0.154	0.086	-0.089	-0.030	-0.110	-0.086	-0.054	-0.077	-0.173	1	-0.220
Anxiety L...	-0.074	-0.494	-0.243	0.351	0.101	0.668	0.189	0.157	0.160	0.518	-0.220	1

Figure 4: Correlation Matrix

- **Box Plots:** To show the correlation between the categorical variables and the anxiety levels we chose to use boxplots.

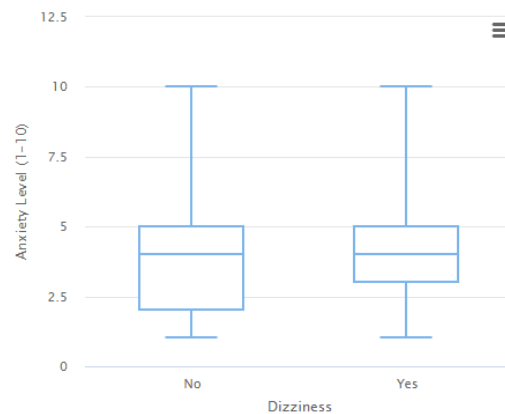


Figure 5: Boxplot of Anxiety levels grouped by Dizziness

We can see from figure 5 that the median anxiety levels are similar for participants with or without dizziness. We can also see that both groups have a wide range of anxiety levels, which may suggest that dizziness alone might not be a strong factor for differentiating anxiety.

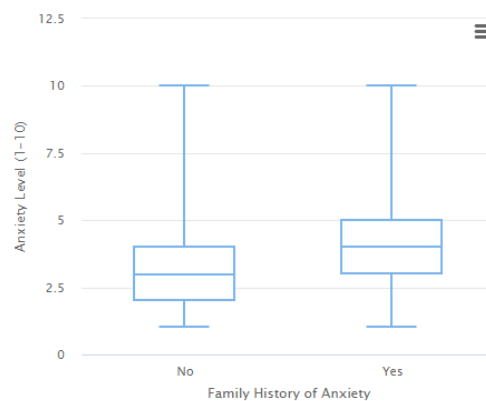


Figure 6: Boxplot of Anxiety levels grouped by Family History of Anxiety

Figure 6 shows that individuals with family history of anxiety have a slightly higher median anxiety levels compared to those without. This may suggest there is a genetic factor which contributes to anxiety levels.

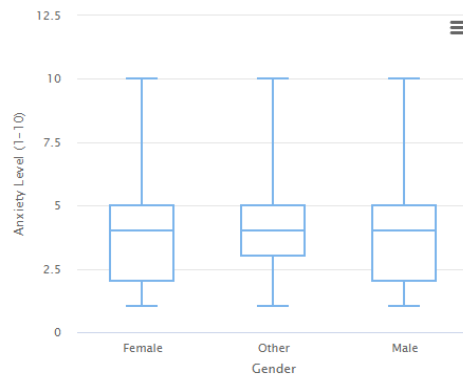


Figure 7: Boxplot of Anxiety levels grouped by Gender

There are very similar median anxiety levels across the genders, as shown in figure 7. However, it is important to note that there may be differences in the spread of anxiety scores, with females showing a very slightly wider distribution.

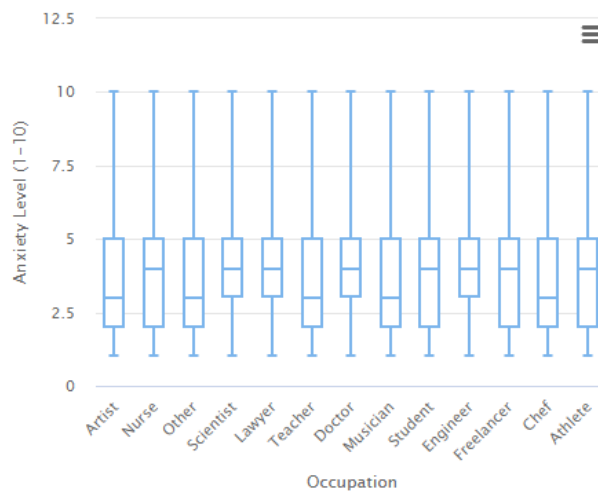


Figure 8: Boxplot of Anxiety levels grouped by Occupation

Across all the occupations available in our dataset, there are not any substantial differences in the reported anxiety levels. The median and spread of anxiety scores are mostly consistent across all occupations. This may suggest that, in this dataset, occupation is not a strong predictor of anxiety levels.

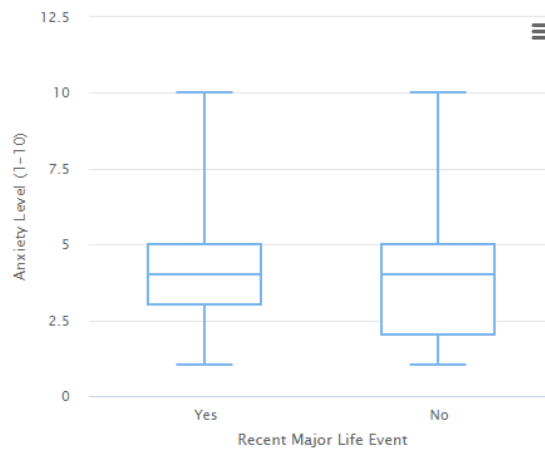


Figure 9: Boxplot of anxiety levels grouped by Recent Major Life Event

As shown in figure 9, those who have experienced a major life event show a slightly higher median anxiety level compared to those who have not experienced any. This points to a possible impact that stressful life altering changes have on anxiety symptoms.

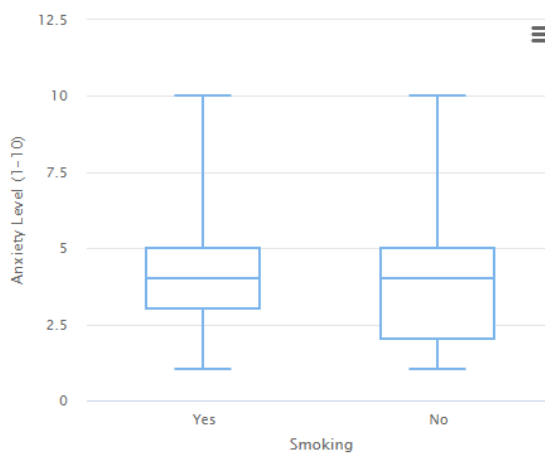


Figure 10: Boxplot of anxiety levels grouped by Smoking

The median anxiety levels between the smokers and the non-smokers are similar, but the smoker group shows a slightly higher median. There would be a need for increased research in this area to draw any deterministic conclusions if smoking has an impact on anxiety levels.

Conclusion:

Across all analyses, *Stress Level*, *Sleep Hours*, *Therapy Session*, *Caffeine Intake*, *Diet Quality*, *Physical Activity*, and *Family History of Anxiety* emerged as the most important predictors of anxiety levels. *Recent Major Life Events* and *Smoking* showed weaker but noticeable associations. Based on their low relevance, *Gender*, *Occupation*, and *Medication* were excluded from further analyses and modeling.

4.2.2 Feature Encoding

After selecting the relevant variables, it was necessary to encode some categorical features to make them suitable for use in machine learning models. We applied **One-Hot Encoding** to transform categorical values into numerical representations.

4.2.3. Feature Transformation

The target variable *Anxiety Level* was initially recorded on a scale from 1 to 10. Although the scale represented levels (categories) of anxiety, classification models interpreted it as numerical data.

We first attempted to directly convert it into a polynomial format, but this resulted in too many categories and led to a very low model accuracy (around 2%). Therefore, we decided to discretize the anxiety levels into three categories: **Low, Medium og High**

This was done using the **Discretize by User Specification** operator in RapidMiner during the modeling process.

4.2.4 Continuous Improvement through Modeling

The feature engineering process was continuously evaluated and adjusted during model testing. Based on the performance of different models, we made necessary changes to the dataset to optimize results for the subsequent steps.

5. Modeling and Evaluation

5.1 Models and validations with split validation

In this phase, we developed models to predict anxiety levels based on the available dataset. Since we transformed Anxiety Level into three categories (*Low*, *Medium*, and *High*), we applied classification models: **Linear Discriminant Analysis (LDA)**, **Decision Tree** and **Neural Network**. The Decision Tree and Neural Network models were not only chosen for their classification performance, but also because they have the ability to identify the most important predictors contributing to anxiety levels.

To evaluate the model performance, we used:

- **Total accuracy** (the percentage of correct predictions)
- **Precision and recall per class** (to assess the model's ability to correctly identify each anxiety level)

Below, we briefly present each model, evaluate the results, and compare performance to determine which model is best suited for this case.

Linear Discriminant Analysis (LDA) and Validation

As discussed in the *feature engineering* phase, Anxiety Level was converted into categorical values. To determine the optimal thresholds, we tested several category groupings. The division into Low (1–3), Medium (4–7), and High (8–10) provided the best balance across classes, enabling the model to learn sufficient variation. This also resulted in higher accuracy, precision, and recall. Other groupings, such as Medium = 4–6, led to poor model performance where only one class was predicted and the model failed to distinguish between levels—resulting in an accuracy as low as 10.32%.

We also tested the effect of including or excluding certain variables. For instance, Medication was initially excluded due to its low weight in the feature selection phase. However, when removed, the model again failed to distinguish between classes and dropped to an accuracy of only 9.23%. Reintroducing this feature restored correct predictions and significantly improved accuracy.

The final LDA model was evaluated using RapidMiner’s Performance operator. It achieved the following results:

accuracy: 79.14%

	true Anxiety low	true Anxiety medium	true Anxiety high	class precision
pred. Anxiety low	846	241	0	77.83%
pred. Anxiety medium	194	713	21	76.83%
pred. Anxiety high	0	3	182	98.38%
class recall	81.35%	74.50%	89.66%	

Figure 11: Confusion matrix for LDA model

These results show that the model is especially effective at identifying individuals with high anxiety—a critical feature in health-related applications. Some overlap between *Low* and *Medium* anxiety was expected, as the boundary between these levels tends to be more fluid in practice.

Decision Tree and Validation

The Decision Tree model was trained using the same set of features and evaluation metrics as used in the LDA model. We also conducted similar tests regarding the category thresholds and the inclusion or exclusion of specific features. Compared to LDA, the Decision Tree model was **less sensitive to changes** in the category definitions.

One of the key advantages of the Decision Tree model is its **visual interpretability**, which allows us to clearly observe how the model makes decisions. The first split in the tree was based on Sleep Hours, followed by Therapy Sessions, and then Stress Level. When compared to the results from the correlation matrix and the “Weight by Information Gain” method, Stress Level was originally the top-ranked predictor. However, the tree confirmed that the same five features were consistently among the most important predictors of anxiety level:

- Sleep Hours
- Therapy Sessions
- Stress Level
- Caffeine Intake
- Diet Quality

The validation of the Decision Tree model yielded results similar to those of the LDA model, particularly in terms of **high recall and precision for the “High Anxiety” class**. The most challenging class to predict accurately was *Medium Anxiety*, which overlapped slightly with both the *Low* and *High* classes. This is understandable, as individuals in the medium range may share characteristics with those in the adjacent levels.

accuracy: 78.09%

	true Anxiety low	true Anxiety Medium	true Anxiety High	class precision
pred. Anxiety low	782	197	0	79.88%
pred. Anxiety Medium	258	746	13	73.35%
pred. Anxiety High	0	14	190	93.14%
class recall	75.19%	77.95%	93.60%	

Figure 12: Confusion matrix for Decision Tree

Neural Network and Validation

To explore whether a more complex model could improve the prediction of anxiety levels, we trained a **neural network** on the same dataset and features used in our other models. Neural networks are well suited for learning **non-linear relationships** and complex patterns in data. Our model consisted of **13 hidden nodes** and **three output nodes**, one for each anxiety category: *Low*, *Medium*, and *High*.

Each input variable (e.g., sleep hours, stress level) is connected to all hidden nodes, with an associated **weight** that represents how strongly that input influences the activation of a node. Likewise, each hidden node connects to all output nodes, again with individual weights indicating how much that node contributes to the probability of each anxiety category.

One node in particular, **Node 2**, stood out due to its strong impact on classification. It had a **high positive weight (9.007)** toward the *Medium Anxiety* output node and a **strong negative weight (−7.960)** toward *High Anxiety*. This means that when Node 2 is activated, it increases the likelihood of the model predicting Medium Anxiety and simultaneously decreases the probability of predicting High Anxiety.

We examined the weights of the input variables into Node 2 and found the most influential to be:

- Sleep Hours: +8.663
- Therapy Sessions: −5.187
- Stress Level: −4.126
- Physical Activity: +4.080
- Caffeine Intake: −2.944
- Diet Quality: +2.635

This combination suggests that Node 2 is activated by individuals with:

- High sleep duration
- High physical activity
- Healthy diet
- Low caffeine intake
- Low stress levels
- Fewer therapy sessions

The model has learned that individuals who activate Node 2 have a high probability of belonging to the Medium Anxiety group, and a low probability of being classified as High Anxiety.

We also analyzed **Node 8**, which had a relatively high positive weight toward *Low Anxiety*.

Interestingly, the same key variables were involved here as well, although the order of importance differed slightly. This reinforces the relevance of these predictors across the model.

For validation, we used RapidMiner's Performance operator. The neural network model showed strong performance, particularly in identifying individuals with **High Anxiety**, achieving a **recall of 98.03%** and **precision of 96.60%** for that class. As with the other models, there was some overlap between *Low* and *Medium Anxiety*, which is expected due to the subtle differences between these categories.

The model achieved an **overall accuracy of 78.73%**, with the following class-level performance:

accuracy: 78.73%

	true Low Anxiety	true Medium Anxiety	true High Anxiety	class precision
pred. Low Anxiety	786	204	0	79.39%
pred. Medium Anxiety	253	747	4	74.40%
pred. High Anxiety	1	6	199	96.60%
class recall	75.58%	78.06%	98.03%	

Figure 13: Confusion matrix for Neural Network

Overall, the Decision Tree model performed well and achieved a **total accuracy of 78.09%**, demonstrating reliable classification capability and useful insights through its interpretable structure.

5.2 Overfitting and Cross Validation

To prevent and detect overfitting, we used 10-Fold Cross Validation.

This method divided the dataset into 10 parts, where the model was trained on 9 parts and tested on the remaining one in each round.

In this way, the model was tested on new and unseen data every time, in contrast to Split Validation, where the dataset is divided only once.

Split Validation can be more susceptible to randomness in the data splitting process and can therefore produce results that vary more when applied to new data.

Cross Validation provides a safer and more robust evaluation of the models.

To achieve a double evaluation, we also applied Cross Validation in addition to Split Validation, which we had already implemented.

This approach gave us a more honest and reliable picture of the models' performance.

Below are the results of the models evaluated with Cross Validation, in the following order: LDA, Neural Network, and Decision Tree.

accuracy: 78.16% +/- 1.15% (micro average: 78.16%)

	true Anxiety low	true Anxiety medium	true Anxiety high	class precision
pred. Anxiety low	4120	1202	0	77.41%
pred. Anxiety medium	1082	3573	109	75.00%
pred. Anxiety high	0	9	905	99.02%
class recall	79.20%	74.69%	89.25%	

Figure 14: Confusion matrix for LDA after cross validation

accuracy: 79.07% +/- 1.31% (micro average: 79.07%)

	true Low Anxiety	true Medium Anxiety	true High Anxiety	class precision
pred. Low Anxiety	3991	1044	0	79.27%
pred. Medium Anxiety	1211	3722	29	75.01%
pred. High Anxiety	0	18	985	98.21%
class recall	76.72%	77.80%	97.14%	

Figure 15: Confusion matrix for Neural network after cross validation

accuracy: 77.96% +/- 1.74% (micro average: 77.96%)

	true Anxiety low	true Anxiety Medium	true Anxiety High	class precision
pred. Anxiety low	3895	1000	0	79.57%
pred. Anxiety Medium	1304	3730	63	73.18%
pred. Anxiety High	3	54	951	94.35%
class recall	74.88%	77.97%	93.79%	

Figure 16: Confusion matrix for Decision tree after cross validation

Additionally, we built a Deep Learning model to further investigate generalization and to avoid overfitting.

We evaluated the Deep Learning model using Cross Validation and obtained the following result:

accuracy: 79.24% +/- 1.16% (micro average: 79.24%)

	true Low Anxiety	true Medium Anxiety	true High Anxiety	class precision
pred. Low Anxiety	3987	1023	0	79.58%
pred. Medium Anxiety	1214	3735	20	75.17%
pred. High Anxiety	1	26	994	97.36%
class recall	76.64%	78.07%	98.03%	

Figure 17: Confusion matrix for Deep Learning model after cross validation

When comparing the three initial models (LDA, Neural Network, and Decision Tree) based on both evaluation methods, LDA initially showed the highest accuracy, with Neural Network being a strong second.

However, after Cross Validation, it became clear that the Neural Network was actually more accurate than LDA.

By running the dataset through the Deep Learning model, we achieved even higher accuracy, especially within the High Anxiety class, which was of particular interest.

The results from both Split Validation and Cross Validation showed only minor differences in accuracy between the two methods.

This indicates that our models generalize well, that the dataset is representative, and that overfitting was not a significant problem in this project.

Nevertheless, we acknowledge that it is never possible to be 100% certain of how models will perform on entirely new and future data.

6. Deployment

Our analysis shows that sleep, caffeine, and physical activity have a clear relationship to Anxiety levels. The lower sleep quality and high caffeine intake result in an increased anxiety level, while high physical activity results in a lower anxiety level. In this analysis, we found that alcohol consumption does not have an effect.

The most influential predictors from the models were sleeping hours, physical activity, caffeine intake, diet quality, therapy hours and stress level. The order of which one of them was the most important varied between the models, but those six were the most central as predictors. Some other factors had little or almost no effect on the anxiety levels, and variables such as gender and jobs were assessed as not relevant to include in the final analysis, which are making the model more practical.

When it comes to the effect of the six factors, the analysis shows that better sleep quality, more physical activity, lower caffeine intake, and a good diet results in a lower anxiety level. On the other side is the high amount of therapy time and increased stress level is associated with higher anxiety level. That is not because therapy increases the anxiety, but rather reflects the persons that seek help

have higher anxiety levels. So lower therapy time and lower stress level is not necessarily a cause of low anxiety, but an indication that the person does not have a high anxiety level.

Still, if a family is known to have individuals with anxiety problems it will result in a higher probability to experience high anxiety levels compared to others who don't have it in their background. Other factors like death or job changes could result in higher anxiety levels but are curable with therapy and medicine use.

Physiological targets such as heart rate, dizziness, and heart rate do have some type of context but are not enough to get a strong prediction alone without any outside factors. The results that we have gathered show that lifestyle, genetic, and outside factors are making it possible to predict anxiety levels in individuals.

6.1 Limitations

Our Results are building upon available variables in the dataset, which means not every individual factor that affects anxiety levels is covered. The anxiety levels are categorized into three to get a better accuracy, but will still mean that the level of detail is low. The models are validated on existing data, and further testing on newer dataset would be important to confirm our findings.

6.2 Conclusion

Our Projects have completed our goals with the help of statistical analysis and machine learning models which have made it possible to identify central factors that could affect anxiety levels such as family history, diet quality, sleep, stress, caffeine intake, and physical activity. Family history was one of the most important factors in the box plots but it was not the most important in the models like the other factors. The models are good at predicting the high anxiety group which is the most vulnerable group. With the findings of the analysis, we can conclude that therapy and medicine will help to decrease the level of anxiety and life stresses. The findings give the data-driven foundations that are useful for further research, especially for the health sector that wants to improve stress prevention and anxiety treatment. Even for those job places that want to show their support for mental health. Our work demonstrates how visualizations and models can make complex relationships more accessible and actionable.

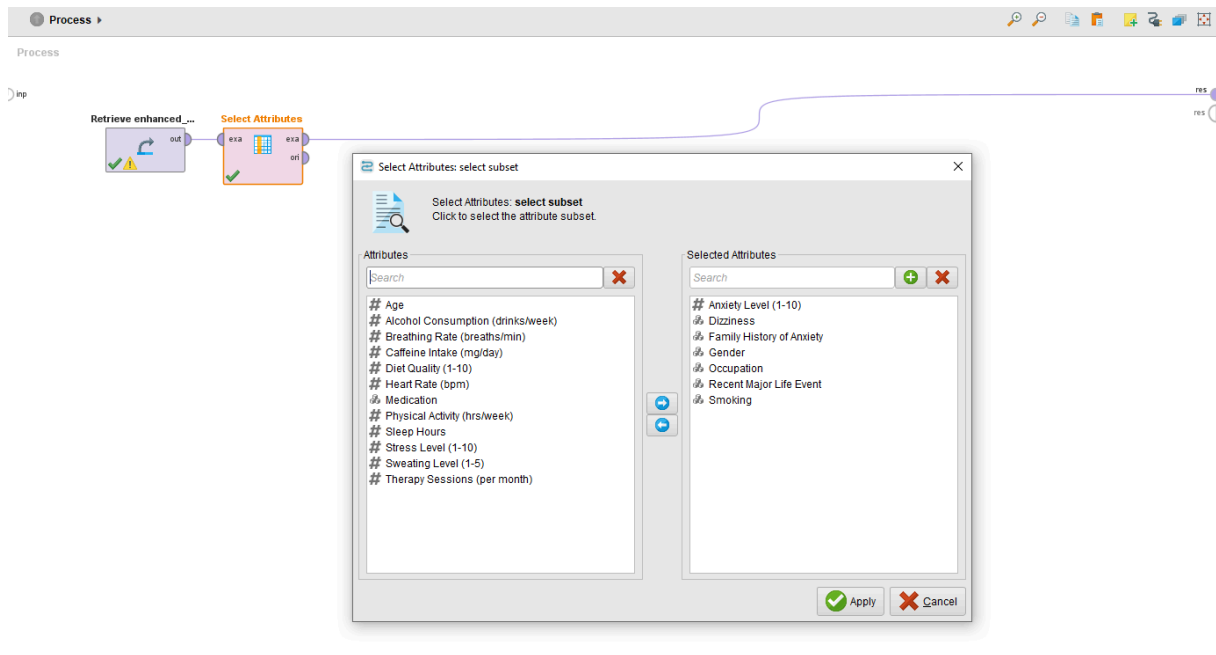
7. References

- Connors, E. (2024, 01.may) *American Adults Express Increasing Anxiousness in Annual Poll; Stress and Sleep are Key Factors Impacting Mental Health*. *Psychiatry.org*. Retrieved April 04, 2025 from <https://www.psychiatry.org/news-room/news-releases/annual-poll-adults-express-increasing-anxiousness>
- Wirth, R., & Hipp, J. (2000). Crisp-dm: Towards a standard process model for data mining. Retrieved April 04, 2025, from <https://cs.unibo.it/~daniilo.montesi/CBD/Beatriz/10.1.1.198.5133.pdf>

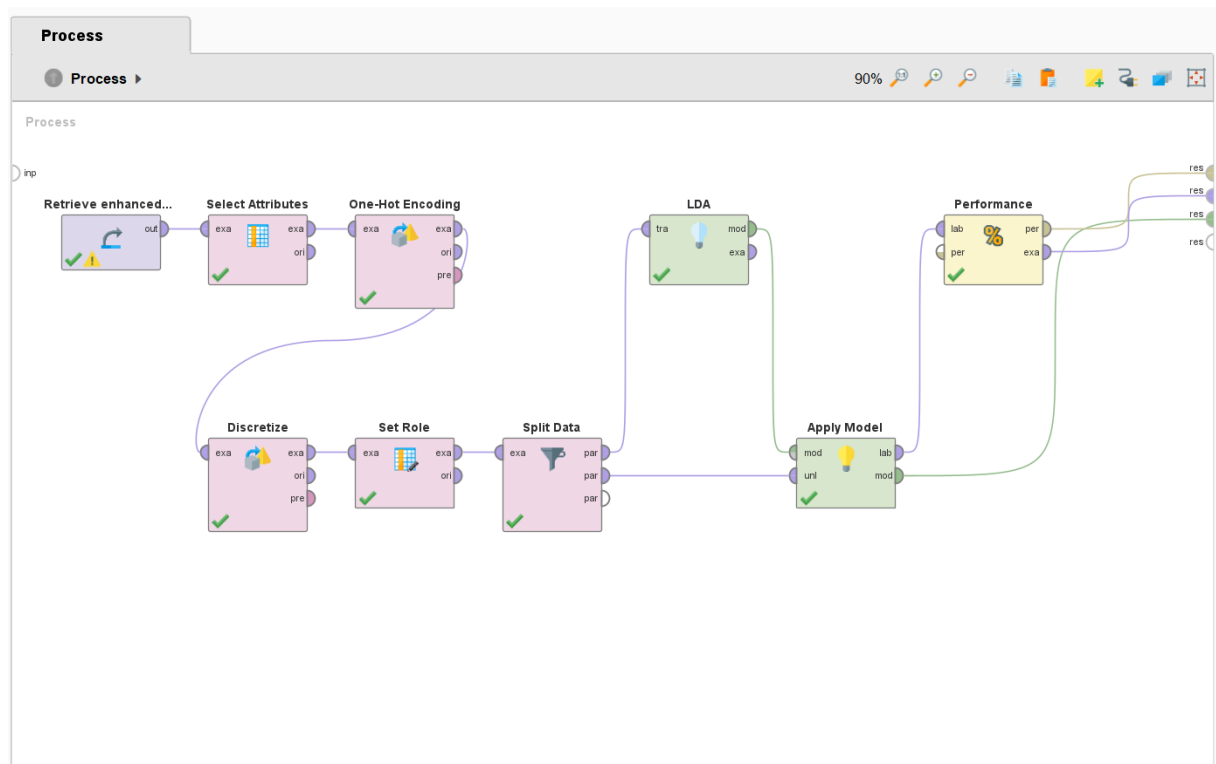
8. Appendices

Group Member Contributions:

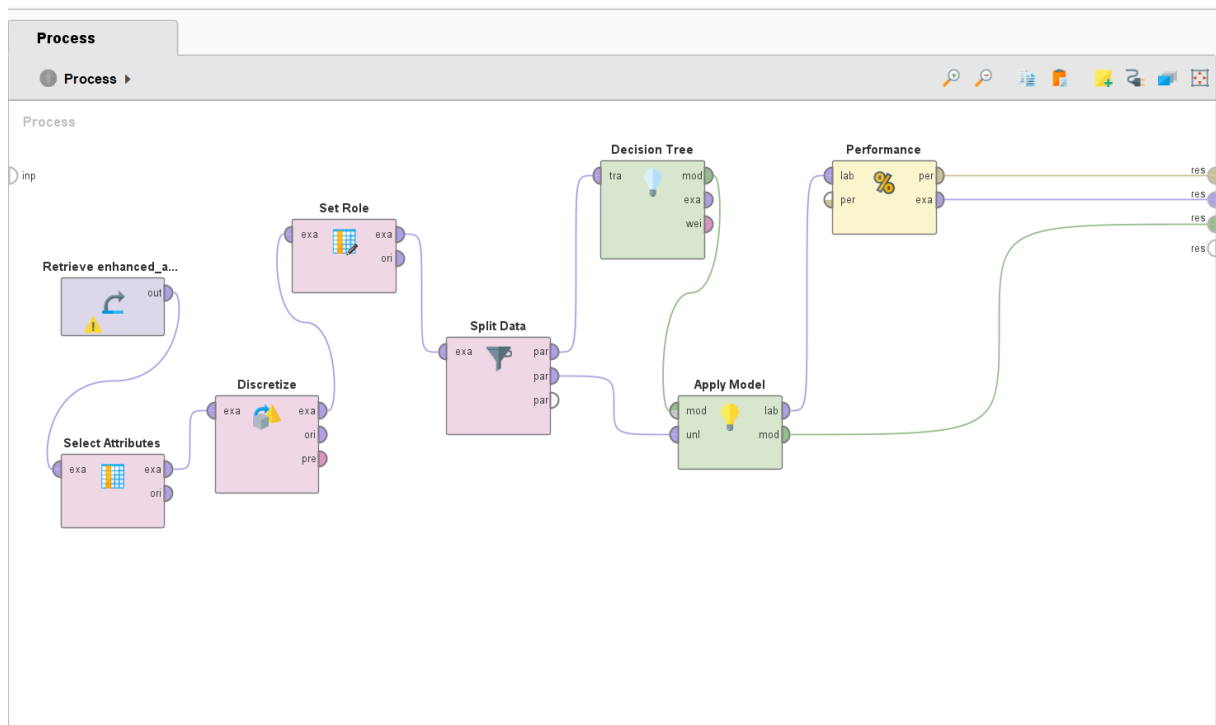
*	Data understanding Data preparation Modelling Evaluation Overfitting
*	Modelling Evaluation
*	Deployment Modelling
*	Data preparation Modelling
*	Introduction, CRISP-DM, Business Understanding, Box Plots, Formatting of report



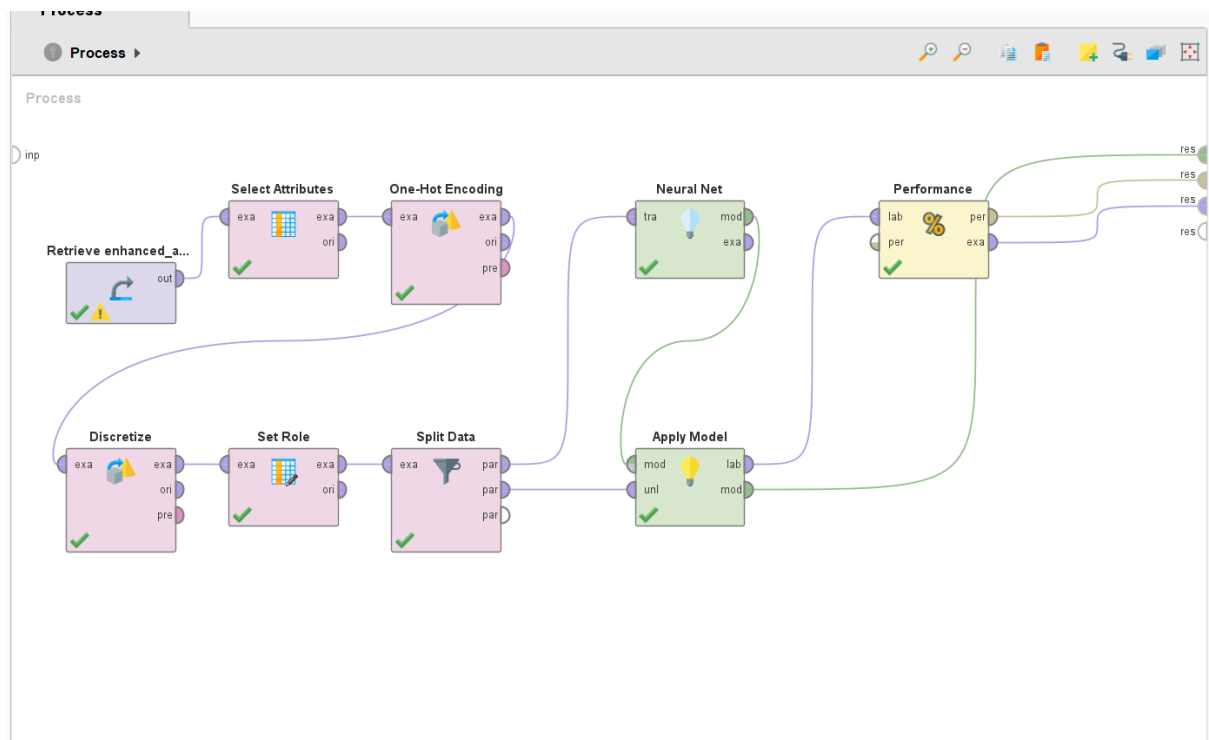
Workflow Description for Boxplots



Workflow description for LDA



Workflow description for Decision Tree



Workflow description for Neural Network

```

Class 'Low Anxiety' (Sigmoid)
-----
Node 1: 0.971
Node 2: 0.779
Node 3: -1.061
Node 4: -0.991
Node 5: 1.106
Node 6: 1.210
Node 7: 1.152
Node 8: 1.505
Node 9: -1.324
Node 10: -1.228
Node 11: -0.868
Node 12: 1.143
Node 13: 0.822
Threshold: -2.787

Class 'Medium Anxiety' (Sigmoid)
-----
Node 1: -0.976
Node 2: 9.007
Node 3: 1.048
Node 4: 0.988
Node 5: -1.102
Node 6: -1.198
Node 7: -1.142
Node 8: -1.499
Node 9: 1.321
Node 10: 1.204
Node 11: 0.876
Node 12: -1.134
Node 13: -0.828
Threshold: -7.003

Class 'High Anxiety' (Sigmoid)
-----

```

Description of Low and Medium Anxiety output nodes for Neural Network model

```

Class 'High Anxiety' (Sigmoid)
-----
Node 1: -0.573
Node 2: -7.960
Node 3: -2.643
Node 4: 2.794
Node 5: 0.356
Node 6: 0.273
Node 7: -1.944
Node 8: -1.173
Node 9: -1.751
Node 10: 4.082
Node 11: -0.517
Node 12: -2.159
Node 13: -1.843
Threshold: 2.569

```

Description of High Anxiety output node for Neural Network model

The screenshot shows a software window titled 'ExampleSet (Apply Model)' with a 'Views' dropdown and a 'Perform' button. The main display area shows the output for 'Node 2 (Sigmoid)' and 'Node 3 (Sigmoid)'. The output for Node 2 lists various input features and their corresponding weights, followed by the bias. The output for Node 3 also lists input features and their weights, followed by the bias.

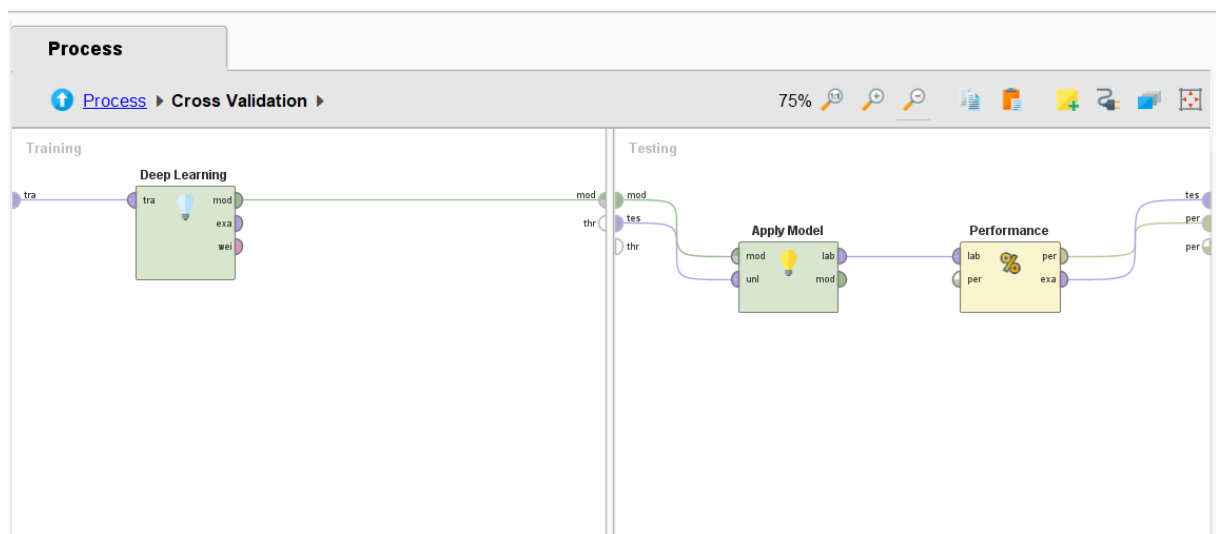
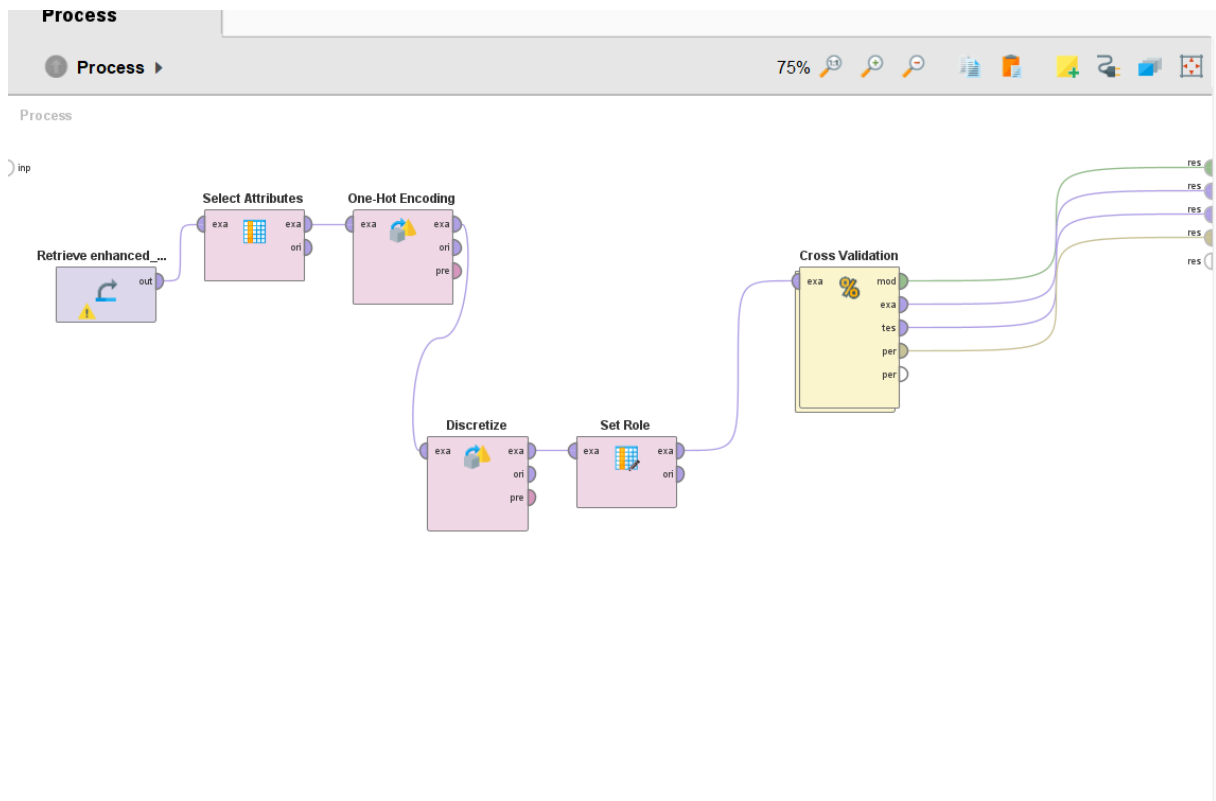
```

Node 2 (Sigmoid)
-----
Smoking = Yes: 0.076
Smoking = No: -0.023
Family History of Anxiety = No: 0.070
Family History of Anxiety = Yes: -0.106
Dizziness = No: 0.432
Dizziness = Yes: -0.412
Medication = Yes: -0.195
Medication = No: 0.157
Recent Major Life Event = Yes: -0.122
Recent Major Life Event = No: 0.102
Age: 1.483
Sleep Hours: 8.663
Physical Activity (hrs/week): 4.080
Caffeine Intake (mg/day): -2.944
Alcohol Consumption (drinks/week): -0.488
Stress Level (1-10): -4.126
Heart Rate (bpm): -2.226
Breathing Rate (breaths/min): -1.306
Sweating Level (1-5): -1.420
Therapy Sessions (per month): -5.187
Diet Quality (1-10): 2.635
Bias: 11.416

Node 3 (Sigmoid)
-----
Smoking = Yes: -2.003
Smoking = No: 1.998
Family History of Anxiety = No: 1.602
Family History of Anxiety = Yes: -1.530
Dizziness = No: -1.629
Dizziness = Yes: 1.718
Medication = Yes: 0.213
Medication = No: -0.304
Recent Major Life Event = Yes: 0.012
Recent Major Life Event = No: -0.046

```

Description of Hidden node 2 that activate Medium Anxiety output node



Workflow for Deep Learning Model with Cross Validation